FEDERAL SPECIFICATION

THIMBLE, EXTRACTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers extraction thimbles, suitable for use with standard extraction apparatus for analytical determination of fat in organic materials.

1.2 **Classification**

1.2.1 *Types and sizes.*—Extraction thimbles shall be of the following types and sizes, as specified (see 6.1):

Type I.—Single thickness wall.
Size 1.—Inside diameter_33 millimeters.
Length _____80 millimeters.
Size 2.—Inside diameter_10 millimeters.
Length _____50 millimeters.

Type II.—Double thickness wall.
Size 3.—Inside diameter_22 millimeters.
Length _____80 millimeters.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU–P–31—Paper; General Specifications and Methods of Testing.

PPP–B–566 — Boxes, Folding, Paperboard.

PPP–B–585—Boxes, Wood, Wirebound.

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL–L–10547 — Liners, Case, Waterproof.

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Material.—Filter thimble shall be fat-free, close-texture filter paper, composed of 100 percent fully bleached cellulose fibers.

3.1.1 *Alpha-cellulose.*—The alpha-cellulose content shall be not less than 95 percent.

3.1.2 *Copper number.*—The copper number shall be not more than 0.5.

3.1.3 *Acidity.*—The pH value shall be not less than 6.0 nor more than 8.0.

3.2 Construction. — Thimble shall be a seamless rigid tube of single or double wall thickness, as specified, with a round bottom. Wall shall be symmetrical, well-compressed and hard, with a smooth inner surface to facilitate removal of extracted matter. The open end shall be cut off squarely and evenly.

3.3 Dimensions.—Extraction thimbles shall conform to the dimensions as shown in table I.

3.4 Workmanship.—Workmanship shall be first class throughout. Thimble shall have uniform formation, and shall be clean and free from imperfections which detract from its appearance or which may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.1). Sampling procedures for the military agencies shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105. In addition, samples may be requested by the procuring activity for testing at a Government laboratory.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or material which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery, as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

TABLE I.

Type	Size	Inside Diameter	Overall length	Wall thickness
		Millimeters	*Millimeters*	*Millimeters*
I	1	33 ± 1	80 ± 2	1 ± 0.3
	2	10 ± 0.4	50 ± 2	1 ± 0.3
II	3	22 ± 0.7	80 ± 2	2 ± 0.4

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4.2.1 *Classification of defects (applicable to military agencies only).*

	Category and defects
Major	*AQL 4 percent defective*
101.	Walls of thimble not free from pinholes.
102.	Walls of thimble not free from cracks, breaks, or other openings.
103.	Walls of thimble not rigid.
104.	Thimble walls not free from dirt particles or other foreign matter (unclean paper pulp used).
105.	Wall surfaces not free from fat, grease, pigments, or other organic or inorganic materials which will tend to adulterate the filtrate.
106.	Dimensions not as specified.
107.	Walls of thimble not seamless.
Minor	*AQL 6.5 percent defective*
201.	Inside diameter and wall thickness not uniform.
202.	Thimble not symmetrical, out of shape.
203.	Open end not evenly or squarely cut.
204.	Thimbles not free from localized thin walls.
205.	Open end not free from chipped, cracked, or fraying edges.
206.	Interior surface not free from overlaps or marked wrinkles.

4.3 Tests.—The following tests shall be performed in accordance with the methods described in Federal Specification UU–P–31:

Tests	*Method No.*
Fiber identification and quantitative determination	300
Alpha cellulose	201
Copper number	220
Acidity, pH	200

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Unit of issue.*—Twenty-five thimbles, as specified, constitute one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE II.—*Packaging quantities*

Federal stock number	Unit package	Intermediate package	Shipping container
6640–444–7850	1 unit	6 units	96 units

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton design shall insure against shifting of, or damage to, the contents. Closure shall be effected in such manner as to prevent spillage of contents under normal handling.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size, or a complete wrapping of kraft paper of suitable weight. Closure shall be effected in such manner as to prevent spillage of contents under normal handling.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

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5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or Military Specification MIL-B-10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or Military Specification MIL-B-10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Civilian agencies.*—Marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Interior packages.*—Each unit package and intermediate package shall be marked in accordance with section 5 of Military Standard MIL-STD-129.

5.3.2.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL-STD-129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and size required (see 1.2.1).

(c) Sampling required (see 4.1).

(d) Inspection required (see 4.2).

(e) Place of final inspection (see 4.2).

(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item appearing in the Armed Services Medical Stock List:

6640-444-7350 THIMBLE, EXTRACTION, 25s

6.3 **Supersession data.**—This specification supersedes Military Specification MIL-E-2280, dated July 13, 1950.

6.4 **Transportation description.**—Transportation description applicable to this item is:

Paper goods, not otherwise indexed by name.

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Carload minimum weight 30,000 pounds.

Motor volume minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C M
Navy—Md
Air Force.

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